



AGENTIC AI

(BAI608A)

Title : AI Based Stampede Crowd Detection System

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Problem Statement

- Monitoring crowd behavior in small or indoor environments (classrooms, corridors, entrances, events) is difficult using manual observation.
- Human supervision cannot continuously track **crowd density and movement patterns** in real time.
- Sudden crowd congestion or panic movement near entry/exit points may lead to **localized stampede situations**.
- There is a lack of simple systems that use **basic hardware like webcams** for intelligent crowd monitoring.



Proposed AI Solution

- Use a **webcam for real-time video capture** to continuously monitor the crowd.
- Apply **Computer Vision techniques (OpenCV)** to process video frames and extract useful information.
- Implement **Human Detection using AI models** (e.g., YOLO, Haar Cascade) to identify people in each frame.
- Perform **Crowd Density Estimation** by counting the number of detected people in a given area.

1. Risk Assessment Agent

Role: Predicts stampede risk

Inputs: Crowd count, Entry/Exit rate, Density

Process:

- Detects abnormal patterns
- Checks imbalance (entry > exit)
- Uses Anomaly Detection

Output: LOW / MEDIUM / HIGH risk

2. Decision Agent

Role: Takes action based on risk

Inputs: Risk level, Crowd state

Process:

- Evaluates severity
- Selects best action

Output (Actions):

- Stop entry
- Open emergency exits
- Send alerts



Agentic AI Cognitive Loop

Perceive → Analyze → Decide → Act → Learn

Perceive: Collect real-time crowd data using the webcam.

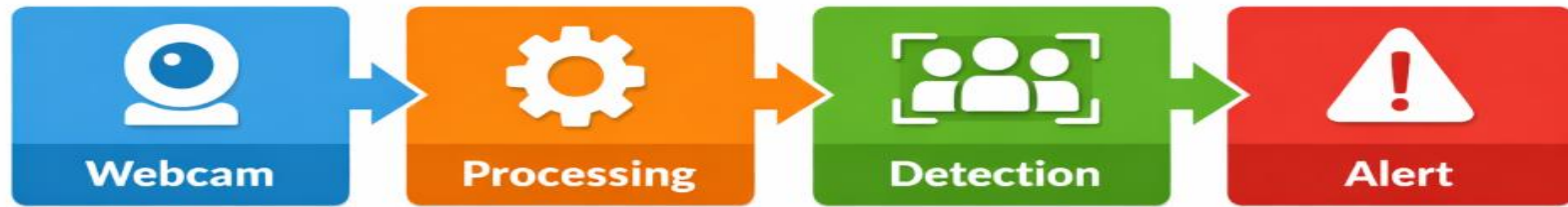
Analyze: Process the data to identify crowd density and movement.

Decide: Evaluate if the situation is safe or risky.

Act: Generate alerts or warnings when risk is detected.

Learn: Update the system using past data to improve accuracy.

System Architecture





Technologies Used

- **Python:** Main programming language for developing the system
- **OpenCV:** For video capture, frame processing, and image analysis
- **YOLO (You Only Look Once):** For real-time human detection.
- **NumPy:** For numerical operations and array processing
- **Webcam:** To capture live video input.
- **Machine Learning / Deep Learning:** For analyzing crowd behavior and patterns.



Key Features

- **Real-Time Monitoring:** Continuously captures and analyzes live video from the webcam Automatic priority & severity detection .
- **Crowd Density Estimation:** Calculates number of people in a given area.
- **Abnormal Behavior Detection:** Detects unusual movements like rushing, pushing, or clustering.
- **Early Warning System:** Generates alerts before a dangerous situation occurs.
- **Low-Cost Setup:** Works using a simple webcam and basic hardware.



- **Integration with CCTV Cameras:** Expand from webcam to large-scale surveillance systems
Map-based complaint tracking (location view) .
- **Drone-Based Monitoring:** Use drones for covering wide and crowded outdoor areas.
- **Advanced AI Models:** Improve accuracy using more powerful deep learning models.
- **Real-Time Notification System:** Send alerts via mobile apps, SMS, or emergency systems.



Conclusion

- The system provides a **low-cost and efficient solution** using a webcam.
- It enables **real-time crowd monitoring** using AI techniques.
- Detects **crowd density and abnormal movements** accurately.
- Helps in **early detection of risky situations**.
- Generates **alerts to prevent stampede incidents**.



Thank You